INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

TIRE, TUBELESS, PNEUMATIC: FOR TRUCK, LOGISTICAL GOER TYPE

This specification is approved for use within the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. Scope

1.1 Scope. This specification covers a single size, tubeless, pneumatic tire for mounting on the M877, M520, M553, and M559 GOER vehicles. The size of the tire is 18.00 - 33 and the ply rating of the tire is 10 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks The followIng specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of SpeclfIcatIons and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinen data which may be of use in improving this document should be addressed to: JS Army Tank-Automotive Command, ATTN. AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document improvement Proposal (DL) Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

UU-T-81	- Tags, Shipping and Stock.
CCC-C-419	- Cloth, Duck, Cotton, Unbleached, Plied-Yarns, Army and Numbered.

MILITARY

MI L-T-4	- Tire, Pneumatic, and Inner Tube. Pneumatic Tire: Tire with Flap; Packaging of.
llIL-P-116	- Preservation, Methods of.
MIL-P-11520	- Preservation Coating, Rubber: For Rubber.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STB721	- Definitions of Terms for Reliability and Maintainability.
MIL-STD-1190	- Minimnm Guidelines for Level C Preservation, Packing and Marking.
MIL-STD-1224	- Visual Inspection Guide for Pneumatic Tires (Non-Aircraft).

2.1.2 Other Government documents, drawings and publications . The following other Government documents drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF TRANSPORTATION (DOT)

Code of Federal Regulations (CFR)

(XR 49	- Transportation.

DRAWINGS

ARMY

11618762	- Tire (Tubeless).

(Copies of specifIcations, standards, drawings and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Non-Government publications The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D412	Rubber Properties in Tension, Test Methods for.
ASTM D746	Brittleness Temperature of Plastics and Elastomers by Impact, Test Method for.
ASTM D1149	Standard Test Method for Rubber Deterioration-Surface Ozone Cracking in a Chamber.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

THE TIRE AND RIM ASSOCIATION

Yearbook.

(Application for copies should be addressed to The Tire and Rim Association, Incorporated, 3200 West Market Street, Akron, OH 44313.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however. shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4.

3.2 Materials Materials shall be as specified herein and as specified in referenced documents (see 4.7.1).

3.2.1 Compound. The tires shall be compounded from natural, synthetic natural rubber (Cls 1-4 polyisoprene), other synthetic rubber or a suitable combination thereof.

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3.2.2 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 6.5.1).

3.3 Design and construction. Tires shall be fabricated as specified herein and in accordance with Drawing 11618762 (see 4.7.1 and 4.7.3).

3.3.1 Carcass. The carcass shall consist of a rubberized cord of material. The carcass may be reinforced with breaker strips to improve impact resistance. Only nylon cord shall be used.

3.3.2 Tread. The tread design shall be directional as shown on Drawing 11618762. - The tread shall be designed to minimize skidding, resist abrasion and protect the carcass from injury.

3.3.2.1 Treadwear indicators. Each tire shall have six treadwear indicators spaced equally around the circumference of the tire.

3.3.3 Sidewall The sidewall shall be designed to protect the carcass against abrasion, and to resist moisture and weatherng-

3.3.4 Bead The bead shall be of such construction as to anchor the tire firmly to the rim without slippage under normal operating conditions and of such design as to fit the standard size 13.00 x 33 rim specified in the Tire and Rim Association-s Yearbook. There shall be no air leakage between the tire and rim.

3-3.5 Tread depth and thickness Tread depth at the center of the tire shall be not less than 1.20 inches and not more than 1.24 inches. The total tread thickness (undertread and tread lug) shall be not less than 1-55 inches

3.3.6 Overwall diameter Tire diameter at the center of the tire tread shall be not less than 70.9 inches and not more than 72.1 inches. Tire diameter after 24 hour inflation growth shall not be greater than 73.2 inches (see 4.7.3-3.1).

3.3.7 Overall tire width. The new tire section width shall be not less than 19.0 inches and not more than 19.6 inches. overall tire width after 24 hour inflation growth shall be not less than 19.0 in and not more than 21.2 inches (see 4-7.3.3.2).

3.3.8 Tire weight Tire weight shall be 417 plus or minus (±) 20 pounds (lbs]. in addition, the weight of tires produced subsequent to first article approval shall not vary by more than ± 3 percent (%) from the weight of the first article sample (See 4.7.4).

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3.4 Mechanical properties (see 4.7.5).

3.4.1 Tensile strength. The tire treads and sidewalls shall have a tensile strength of not less than 2300 pounds per square inch (psi).

3.4.2 Elongation. The tire treads and sidewalls shall have an ultimate elongation of not less than 500% (2 to 12 inches).

3.4.2.1 Modulus of elasticity. At the time tensile strength and ultimate elongation are being determined, the modulus (stress tensile) at 300% elongation shall be recorded.

3.5 Performance.

3.5.1 General Tires shall retain air under the pressure required to support the GOER type vehicles loaded with rated payloads and operated in accordance with the vehicles' opeational profile (see 4.7.1 and 6.4).

3.5.2 Indoor endurance. There shall be no evidence of tire damage after 42 hours of operation at 100 ± 5 degrees Fahrenheit (°F) and a speed of 35 miles per hour (mph) while supporting loads varying from 5750 to 8200 lbs (see 4.7.6.1).

3.5.3 Environmental.

3.5.3.1 Temperature.

3.5.3.1.1 Storage. Unmounted tires shall withstand extended storage for periods up to 66 months in ambient air temperatures ranging from plus 155 to minus 65°F without deterioration (see 4.7.1).

3.5.3.1.2 Operation. Tires shall have the inherent capability of acceptable performance in ambient air temperatures ranging from plus 125 to minus 65°F (see 4.7.1).

3.5.3.1.3 Low temperature brittleness. When tested at temperatures as low as minus 65 ± 2°F, tire specimens shall evidence no crazing, cracking, breaking, separation or rupture (see 4.7.6.2.1).

3.5.3.2 Ozone resistance. The specimens taken from tires shall be ozone resistant equal to or better than that of the Government control compound specified in paragraph 4.7.6.2.2.1.1.1. If any cracking becomes evident in any of the test specimens during the tests, it shall not appear faster than, nor shall the degree of cracking be greater than, that of the Government control specimen. (see 4.7.6.2.2 and 6.3).

3.5.4 Road service. When specified (see 6.2), a road service test shall be conducted by the Government. After the road service test, tires shall show no evidence of broken cords, separation of tread, ply, cord or bead, or of cracking in the tread or sidewalls sufficient to expose the fabric. There shall be no fabric visible as a result of tread wear (see 4.7.6.3).

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3.6 Age Of tires Tires submitted for shipment under this specification shall not be more than 12 months old in age from the date of manufacture. Tires exceeding 12 months from the date of manufacture may be submitted for shipment providing they are prepared in accordance with 5.1.1.1, 5.1-1.2 and 5.1.1.3 (see 4.7.7).

3.7 Marking. Tires shall be marked in accordance with DOD or DOT requirements. DOD marking requires the manufacturer's name or trademark. nominal tire size, ply rating, and identification or serial number shall be branded on each tire. The competition of the ply cord (for example. "nylon" shall be identified permanently and legibly on the sidewall of the tire. Tires shall be branded with the word "Tubeless". The month and year of manufacture shall be branded or permanently shown on the serial side of each tire. This marking shall be near the serial number and of a size comparable thereto. Tires exempt from DOD marking requirements shall be marked in the same manner as required by DOT Part 574 of title 49 CFR, using the first two and last three digits of the code for identification, except zero (0) shall fill the remaining places (see 4.7.2).

3.8 Workmanship Workmanship shall be such that the tires meet the requirements of this specification. Defects, which may impair the serviceability of tires, the specific limits of these defects, and the permissible corrective actions are detailed herein (see 4.7.2).

3.8.1 Cord buckles. Cord fabric shall not contain folds or buckles which cause any part of a given ply, except splices, to be out of its proper plane.

3.8-2 Insufficient ply coating. . Ply coating shall meet the requirements of 3.3.

3.8.3 Bead kink Beads shall not be kinked.

3.8.4 Sidewall cracks or folds Tires shall be free from sidewall cracks or folds- Any such cracks or folds which occur in production must be removed. If a crack or fold is of a depth greater than one-half of the thickness of the sidewall cover stock at the location of the crack. then it must be eliminated by a Vulcanized spot repair which restores the area to its required thickness. if a crack or fold is of a depth up to one-half of the sidewall thickness at the location of the crack, then it may be elimimated by buffing to a smooth surface with the general tire contour preserved. In neither case should the repair or buffing operation damage the cords beneath the sidewall stock.

3.8.5 Ply separation. Tires shall be free from ply separation.

3.8.6 Thread folds. Tires shall be free from tread folds.

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3.8.7 Sidewall folds Tires shall be free from sidewall cover stock folds in the sidewall area. Folds shall be repaired when not longer than 6 inches in one location.

3.8.8 Complete marking. Marking shall be complete, conforming to the requirements specified in 3.7. Marking may be restored or added with impressed or raised characters, which approximate the same size as those produced by the mold.

3.8.9 Thread crates and voids. Tires shall be free from tread craters more than 3 inches in length and free of voids. Tread craters shorter than 3 inches in length, and less than one-half the skid depth or element width shall be spot repaired in one location per tire.

3.8.10 Open tread splice. Tires shall be free from open tread splices or depressions not more than 1 inch or deeper than 15% of the skid depth. Open tread splices or depressions less than 1 inch in length or less-than 15% of the skid depth may be repaired.

3.8.11 Open sidewall cover splice. Tires shall be free from sidewall cover splices. Any such open sidewall cover splices which occur during production and extend to a depth of one-half of the sidewall cover thickness shall be repaired provided the repairs are limited to one per sidewall.

3.8.12 Sidewall cove blisters. Tires shall be free from sidewall cover blisters. Any such blisters which occur during production and are not more than 1 square inch in area shall be repaired provided the tire does not have more than three per sidewall and shall be more than 6 inches apart.

3.8.13 Sidewall cover blemishes. Tires shall be free from sidewall cover blemishes. Blemishes occuring during production shall be removed. If a blemish is of a depth more than one-half of the sidewall cover stock thickness, then it shall be eliminated by a vulcanized repair which restores the area to its required thickness. If a blemish extends to a depth up to one-half of the sidewall cover stock thickness; it may be removed by buffing to a smooth surface with the general tire contour preserved.

3.8.14 Bead trimming. When bead trimming is required, the trimming shall not expose the fabric.

3.8.15 Narrow Tires shall be free from too narrow beads not meeting requirements.

3.8.16 Bead roughness Tires shall be free from bead roughness. Roughness occurring during production shall be smoothed by triming or buffing providing fabric is not exposed.

3.8.17 Bead fabric Tires shall be free from loose or torn bead fabric.

3.8.18 Exposed cords inside. Tires shall be free from exposed cords inside.

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3.8.19 Tread-element rounding. Tires shall be free from tread-element rounding..

3.8.20 Tread-element edges. Tires shall be free from chipped. rough, or torn tread edges.

3.8.21 Tread Tires shall be free from tread blows.

3.8.22 Sharp edges and airbag roughness Tire interior shall be free from sharp edges and airbag roughness. Airbag venting grooves are allowable provided the grooves have no sharp edges.

3.8.23 Foreign Tires shall be free from foreign material cured into the tire.

3.8.24 Wavy cords Tires shall be free from wavy cords.

3.8.25 Loose tuck-under. Tires shall be free from loose tuck-under.

3.8.26 Mold tearing. Tires shall be free from mold tearing (parting of tread stock during removal of tire from mold).

3.8.27 Oxidized liner stock. Tires shall be free from oxidized (surface aged) liner stock .

3.8.28 Liner splice opening. Tires shall be free from liner splice openings.

3.8.29 Off-register treads. Tires shall be free from tread diametrically off-register and from mold halves in angular misalignments.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order (see 6.2), the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein. unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth n the specification where such inspections are deemed necessary to assure supplies and service conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of

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assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections:

a. First article inspection (see 4.4).
b. Quality conformance inspections (QPI) (see 4.5).
   1. Examination (see 4.5.2).
   2. Tests (see 4.5.3).
c. Control tests (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following standard (room) ambient conditions:

a. Temperature: 77 ± 18°F.
b. Relative humidity: Uncontrolled room ambient.
c. Atmospheric pressure: Site pressure.

4.4 First article inspection. When specified (see 6.2), the Government shall select two tires produced under the production contract for first article inspection. First article tires shall be inspected as specified in table I.

TABLE I. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance		Control
				Examination	Tests	
Group A:						
Materials and construction	3.2, 3.3, 3.5.1, 3.5.3.1.1 and 3.5.3.1.2	4.7.1	X			
Defects (see table II)	3.7, and 3.8	4.7.2	X	X		X
Group B (Performance):						
Physical dimensions	3.3, 3.3.5, 3.3.6 and 3.3.7	4.7.3	X			X

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TABLE I. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance		
				Examination	Tests	Control
Tire weight	3.3.8	4.7.4	X		X	X
Mechanical properties 1/	3.4	4.7.5	X			X
Indoor endurance	3.5.2	4.7.6.1	X			X
Group C (Environmental):						
Low temperature brittleness 1/	3.5.3.1.3	4.7.6.2.1	X			
Ozone resistance 1/	3.5.3.2	4.7.6.2.2	X			X
Group D (Life):						
Road service	3.5.4	4.7.6.3	X			
Age of tires	3.6	4.7.7			X	

1/ Tests shall be performed on certified sheet stock representative of the tire being inspected.

4.5 QCI. QCI shall include the examination of 4.5.1 and the tests of 4.5.2 (see 6.2). Noncompliance with any of the specified requirements in sections 3 and 5 shall be cause for rejection.

4.5.1 Examination. Tires shall be examined for the defects specified in table II. Defects shall be determined where applicable in accordance with typical photos of MIL-STD-1224.

4.5.2 QCI tests. Tires shall be subjected to the QCI tests specified in table I.

4.6 Control tests. Tires selected for control tests shall be new and subjected to the control tests specified in table I (see 6.2).

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TABLE II. Classification of defects.

Category		Defect	Method of examination
Critical		None	
Major			
	101	Cord buckles (see 3.8.1).	Visual
	102	Insufficient ply coating (see 3.8.2).	SIE 1/
	103	Bead kink (see 3.8.3).	SIE
	104	Sidewall cracks or folds (see 3.8.4).	SIE
	105	Ply separation (see 3.8.5).	Visual
	106	Tread folds (see 3.8.6).	Visual
	107	Sidewall folds (see 3-8.7).	SIE
	108	Tread craters and voids (see 3-8.9).	SIE
	109	Open tread splice (see 3.8.10)	SIE
	110	Open sidewall cover splice (see 3.8.11).	SIE
	111	Sidewall cover blisters (see 3.8.12).	SIE
	112	Sidewall cover blemishes (see 3.8.13).	SIE
	113	Bead trimming (see 3.8.14).	Visual
	114	Narrow beads (see 3.8.15).	515
	115	Bead roughness (see 3.8.16).	Visual
	116	Bead fabric (see 3.8.17).	Visual
	117	Exposed cords inside (see 3.8.18).	Visual
	118	Tread-element rounding (see 3.8.19).	Visual
	119	Tread-element edges (see 3.8.20).	Visual
	120	Tread blows (see 3.8.21).	Visual
	121	Sharp edges and airbag roughness (see 3.8.22).	Visual
	122	Foreign material (see 3.8.23).	Visual
	123	Wavy cords (see 3.8.24).	Visual
	124	Loose tuck-under (see 3-8.25).	Visual
	125	Mold tearing (see 3.8.26).	Visual
	126	Oxidized liner stock (see 3.8.27).	Visual
	127	Liner splice opening (see 3.8.28).	Visual
	128	Off-register treads (see 3.8.29).	SIE
Minor			
	201	Incomplete marking (see 3-7 and 3.8.8).	Visual
	202	Sidewall cracks; shallower than one-half of wall thickness (see 3.8.4).	SIE
	203.	Open tread splice: 1 inch or less in length and 15% or less of skid depth, and with no depression under splice area (see 3.B.l0)-	SIE
	204	Open sidewall splice only one splice per tire and splice depth one-half or less of sidewall thickness, and with no depression under splice (see 3.8.11).	SIE

1/ SIE = Standard Inspection Equipment.

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4.7 Methods of inspection.

4.7.1 Design and construction. Conformance to 3.2, 3.3, 3.5.1, 3.5.3.1.1 and 3.5.3.1.2 shall be determined by inspection of contractor records providing proof or certification that design, construction. processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records. processing and quality control standards, vendor catalogs and certifications, industry standards, test reports and rating data.

4.7.2 Defects Conformance to 3.7 and 3.8 shall be determined by examination for the defects listed in table II. Examination shall be visual or by measurement with standard inspection equipment.

4.7.3 Physical dimensions.

4.7.3.1 Skid depth. To determine conformance to 3.3, the skid depth shall be measured and compared to the dimension specified on the drawing.

4.7.3.2 Tread contour. To determine conformance to 3.3, the tread contour shall be exanined and compared to the drawing.

4.7.3.3 Mounted tire. To determine conformance to 3.3, the tire shall be mounted using the rim specified in 3.3.4 and inflated to 30 psi. The tire shall be allowed to stand for 24 hours at room temperature and then adjusted to the specified pressure. With the tire thus prepared, the following measurements shall be made.

4.7.3.3.1 Overall diameter To determine conformance to 3.3.6, the outside circumference of the tire shall be measured to the nearest tenth of an inch and the result divided by 3.1416 to determine the diameter.

4.7.3.3.2 Overall tire width. To determine conformance to 3.3.7, the tire's overall width shall be calipered at six different locations, spaced equally around the circumference, and the average of these measurements determined. The measurements shall include the width of side ribs bars or decorations if they contribute to overall width.

4.7.4 Tire weitht

4.7.4.1 First article inspection. To determine conformance to 3.3.8, the tire shall be weighed.

4.7.4.2 Quality conformance and control inspections. To determine conformance to 3.3.8, the tire shall be weighed and the weight shall be compared to the weight of the first article sample.

4.7.5 Mechanical properties. To determint conformance to 2.4, samples taken from sheet stock certified as being the same compound and state of cure as the material used in the tread and sidewalls of the tire being supplied on

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the contract shall be subjected to tests for tensile strength and elongation in accordance with ASTM D412. The stress at 300% elongation shall be recorded.

4.7.6 Performance

4.7.6.1 Indoor endurance To determine conformance to 3.5.2, the tire shall be mounted on frim specified in 3.3.4, inflated to 35 psi and subjected to the test specified herein under the conditions specified in table 111.

TABLE III. Conditions for the indoor endurance test.

t

Period	Percent of the load	Time (hours)	Speed (mph)	Test load (lbs) 1/
1	70	7	30	5,750
2	90	16	30	7,400
3	100	24	30	8,200

1/ Test loads were determined by multiplying the percent of tire load by 88% of a rated load of 9,340 lbs.

4.7.6-1.1 Apparatus. The apparatus shall include a flat-faced steel wheel 67.23 inches in diameter (1/300 mile in circumference) and at least as wide as the cross section of the tire to be tested. Testing shall be conducted at a controlled ambient temperature of 100 f 5±F.

4.7.6.1.2 Procedure The tire, mounted and inflated as required, shall be pressed against the test wheel with the initial test load specified in table III. Then with the ambient air held at a temperature of 100° ±5°F, the tire shall be run at each of the test loads, for the time and at the speed shown in table III. At the end of each period, a record shall be made of the Inflation pressure, with the first reading taken 3 hours after start of the test. In the event the pressure falls below the first value at later readings, it shall be brought up to the first value while the tire is at running temperature.

4.7.6.2 Enviromental

4.7-6.2-1 Low temperature brittleness. To determine conformance to 3-5.3.1.3, tread and sidewall specimens shall be taken from sheet stock certified as being of the same compound and state of cure as the tread and sidewall of the tire being supplied on the contract. The size of the samples and the method of testing shall conform to ASTM D746 except that the sample width shall be 0.075 inch. Testing shall be conducted at minus 65 ± 2°F Upon completion of the test, the specimens shall be examined.

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4.7.6.2.2 Ozone resistance. To determine conformance to 3.5.3.2, three test specimens of the formulated compound as noted in table IV, measuring 3.075 X 0.010 X .075 in to 0.10 in shall be thoroughly cleaned using methyl alcohol and conditioned for 24 ± 1 hours at 73 ± 4F and 50 ° 5% relative humidity (RH). Test specimens shall then be elongated 25 percent and placed in an ozone chamber in accordance with ASTM D-1149. Accelerated ozone exposure tests shall be conducted in accordance with ASTM D1149 at a concentration of 50 parts per hundred million (pphm) ozone to air at 100 t 2LJF for a period of 72 hrs. Specimens shall then be removed and checked

4.7.6.2.2.1 Dynamic ozone test

4.7.6.2.2.1.1 Test specimens. A total of five specimens shall be used for dynamic testing. Four specimens shall be from certified sheet stock, and one shall be a government control specimen. Two samples for each test shall be sidewall stock.

4.7.6 .2.2-1.1.1 Government The Government control specimen shall be taken from sheets compounded from the formula specified in table IV and cured 60 minutes at 287~F without mold lubricants. The control specimen shall be 1 by 4 inches and shall be new and untested. The mill grain of the control specimen shall be parallel to the 4 inch length. The mill grain of the control specimen shall be buffed on one side to a thickness of 0.075 ± 0.005 inch.

TABLE IV. Formula for Government control compound used in ozone tests.

Materials	Parts by weight
SBR 1500	100.0
HAF Type Carbon Black (N330)	50.0
ZnO	3.0
Stearic Acid	1.5
Highly Aromatic Proc. Oil	7.0
Sunolite 240 (WHITCO)	1.5
Santoflex 13	3.5
Sulfur	2.0
Tertiary-Butyl Benzothiazole Sulfenamide	1.25 (TBBS)

4.7.6.2.2.1.1.2 Tire tread and sidewall test specimens Tread and sidewall specimens shall be taken from sheet stock certified as being of the same compound and state of cure as the material used in the tread and sidewall of the tires being supplied on t-he contract. The test specimens shall be new and previously untested and shall be 1 inch in width and as near to 4 inches in length as possible. The test specimens shall be buffed. if necessary, to a thickness of 0.075 ± 0.005 inch and cemented to the specimen belt with the length of the test specimen parallel to the length of the belt.

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4.7.6.2.2.1.2 Apparatus. The testing apparatus shall include an ozone test chamber conforming in general to that shown in figure 1. The source of ozone shall be Hanovia No. '2851 Safe-T-Air lamp with quartz envelope, or equivalent. Power for the lamp shall be supplied by a 3000-volt Sola transformer, or equivalent. An exhaust blower of 60 cubic feet per minute capacity shall be installed at the top of the chamber. The blower shall be fitted with a damper to control airflow, thus controlling the ozone concentrate ion. The chamber shall be equipped with two 2 l/2-inch diameter pulleys, with the upper pulley powered to rotate at 350 revolutions per minute (rPm). The upper pulley shaft shall be allowed to move up or down in slots in the sides of the chamber. The lower pulley shaft shall have a 40-lb weight suspended from it to maintain sufficient belt tension to achieve maximum flexing of the specimens around each pulley. The specimen belt, on which the specimens are cemented, shall be of cotton duck conforming to type I cloth, cotton duck, unbleached plied yarns of CCC-C-419, or their equivalent. The belt shall be 4 inches wide and 69 inches long. This size belt can accommodate up to 26 specimens. The speed of the belt shall be approximately 40 rpm.

4.7.6.2.2.1.3 Procedure Each group of specimens, tested at one time and place, shall include at least one Government control specimen. The belt. with the specimens cemented thereon, shall be installed on the pulleys in the chamber. Air, with a relative humidity of 45 to 55% and at a temperature of 750 ± 5oF, shall be drawn into the chamber by the blower. The damper on the blower shall be regulated so that the concentration of ozone in the chamber will remain at 30 ± 5 parts of ozone per 100,000,000 parts of air by volume. The belt shall be operated and the specimens exposed for 48 hours. During the 48-hour period, the specimens shall be observed (without magnification) after 24 and 48 hours of exposure.

4.7.6.2.2.2 Static ozone test.

4.7.6.2.2.2.1 Test specimens. A total of five specimens shall be used for static testing. Four specimens shall be from certified sheet stock and one shall be a government control specimen. Two samples for each test shall be sidewall stock.

4.7.6.2.2.2.1-1 Government control specimen. The Government control specimen shall be taken from the test sheets compounded from the formula specified in table IV and cured for 60 minutes at 2570F without mold lubricant- The specimen shall be 1 inch in width and of such length to provide, as nearly as possible, 3 inches of unstressed exposed length between the clamps of the specimen holder shown in figure 2.

4.7.6.2.2.1.2 Tire tread and sidewall test specimens . The tire tread and sidewall test specimens shall be taken from sheet stock certified as being of the same compound and state of cure as the material used in the tread and sidewall of the tire supplied on the contract. The test specimens. which shall be new and previously untested, shall be 1 inch wide and of such

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length as to provide, as nearly as possible, 3 inches of unstressed exposed length between the clamps of the specimen holder shown in figure 2. The test specimens shall be buffed. if necessary, to a thickness of 0.075 ± 0.005 inches.

4.7.6.2.2.2.2 Apparatus. The apparatus shall include an insulated test chamber, with an ozone generating source outside the chamber. Means shall be provided for measuring the ozone concentration, for controlling the temperature of the am in the chamber, and for circulating the air. The apparatus shall also include specimen holder as shown in figure 2.

4.7.6.2.2.2.3 Procedure. Each group of specimens, tested at one time and place, shall include at least one Government control specimen. Bench marks, 2 inches apart, shall be placed on each specimen. The specimen, while straight but unstressed, shall be clamped in the holders with the clamps set as close to 3 inches apart as the length of the specimen will allow. The portion of the specimen between the bench marks shall be centered between the clamps. The specimens shall then be stressed by adjusting the clamps until the 2-inch space between the marks is elongated to 2 1/4 inches (12 1/2%). The stressed specimens, while elongated in the clamps, shall be conditioned at room temperature for 45 ± 5 minutes, and then exposed for 7 days in the test chamber in which air throughout the test, having an ozone concentration of 50 ± 5 parts of ozone per 100,000,000 parts of air by volume, is circulated at a temperature of 100 ± 5oF. If a specimen breaks at or near where clamped, it shall be reclamped and restressed to proper length. The surfaces of the portion of the specimens between the bench marks shall be examined frequently on the molded surface sides under 9-power magnification.

4.7.6.3 Road service. To determine conformance to 3.5.4. the road test shall consist of 100 miles of operation under the vehicle specification conditions of load- Speed, terrain, and environment shall be based on the vehicles mission profile in accordance with table VI (see 6.4).

4.7.7 Age of tires To determing conformance to 3.6, tires shall be examined to assure that tire age is not in excess of time specified.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Cleaning. Tires shall be cleaned in accordance with MIL-P-116.

5.1.1.2 Drying Tires shall be dryed in accordance with MIL-F-116.

5.1.1.3 Preservation. Tires shall be preserved with a penetrating preservative in accordance with MIL-P-11520.

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5.1.2 Level B The level B preservation shall be the same as level A.

5.1.3 Level C The level C preservation shall be in accordance with MIL-STD-1190.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A and B. Tires preserved as specified in 5.1, shall be packed in accordance with MIL-T-4.

5.2.2 Level C. Tires preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1190.

5.3 Marking. Tires shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. Tires covered by this specification are intended primarily for mounting on "GOER" type vehicles.

6.2 Ordering data Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. If first article is required (see 3.1).
d. If road service is required (see 3.5.4).
e. If DOD or DOT marking is required (see 3.7).
f. If responsibility of inspection and place of Inspection should be other than specified (see 4.1).
g. If inspection conditions shall should be other than specified (see 4.3).
h. Types of first article sample, the number of samples to be inspected (see 4.4 and 6.6).
i. QCI acceptance criteria (see 4.5 and 6.7.1).
j. Control test sample size, frequency of tests, and disposition of lots covered by the sample size and of defective items (see 4.6 and 6.7.2).
k. Selection of applicable packaging requirements (see 5.1 and 5.2).

6.3 Government control specimen. The Government control specimen is a special compound made and used by the Government in the ozone tests as a control 1 or comparison specimen for determining the ozone resistance qualities of the manufactuer's product. The manufacturer may at

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any time make and use the compound for test purposes in evaluating ozone resistance of his own compounds. The specimens are compounded in accordance with the formula specified in table IV and cured as specified in 4.7.6.2.2.1.1.1 and 4.7.6.2.2.2.1.1. No mold lubricant is used during the process.

6.4 Typical vehicle operational profiles. Typical vehicle operational profiles are given in table V. One mission shall consist of operation under the conditions specified in table VI. Vehicle loads will comply with vehicle specification requirements.

TABLE V. Typical vehicle operational profiles.

Model	Vehicle renewal life (miles)	Average speed (mph)	Mission duration (miles) 1
M520/M877	25,000	20.0	100
M559	18.000	14.4	100
M553	12,500	10.0	!50

TABLE VI. Mission conditions. 1/

Terrain	% Distance (% of mission duration)	Speed range (mph)	Operation time (minutes)
Hard Surface	~0	0-30	
Secondary	40	0-30	
Level Cross Country	16	0-30	
Level Cross Country (mud) 2/	4	0-10	
Hilly Cross tiuntry u	20	0-15	
Swimmng v	traverse barrier		5

1/ The cllmatic extremes for temperature and altltude shall be minus 25 to 125°F and 500 to 5,000 feet respectively.
2/ 4-wheel drive as required.

6.5 Definitions.

6.5.1 Recover ed materials "'Recovered materials" means materials that have been collected or recovered from solid waste (see 6.5.2).

6.5.2 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material. including solid, liquid. semisolid, or contained gaseous material resulting from

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industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations. section 23.402).

6.6 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the first article sample(s) should be a preproduction sample an initial production sample, a first production item or a standard production item from the contrator's current inventory; the-number of samples to be inspected as specified in 4.4; and (when applicable) the specific tests to be performed on each sample. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the-bid that prior Government approval is presently appropriate for the pending contract- Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.7 Inspection. The following information is presented as guidance for contracting officer in specifying sampling and acceptance procedures for and control testing in acquisition documents.

6.7.1 QCI.

6.7.1.1 Sampling

6.7.1.1.1 Lot formation. An inspection lot should consist of all the tires of one part number, from an Identifiable production period, from one manufacturer, submitted at one time for acceptance.

6.7.1.1.2 Sampling for examination. Samples for quality conformance examination should be selected in accordance with general inspection level II of MIL-STD-105.

6.7.1.1.3 Sampling for tests. Samples for quality conformance testing should be selected in accordance with inspection level S-3 of MIL-STD-105.

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6.7.1.2 Examination.

6.7.1.2.2 Acceptable quality level (AQL). Each sample selected in accordance with 6.7.1.1.2 should be examined to determine conformance to the following AQLs.

Classification	AQL
Major	2.5
Minor	4.0

6.7.1.3 Tests. Samples selected in accordance with 6.7.1.1.3 should be subjected to tests specified in table I.

6.7.2 Control tests.

6.7.2.1 Control test sample. The control test sample should consist of one tire. Samples should be selected at random at the following rates:

Calendar-year tire production (per plant)	Number of control test samples (per calendar year)
1-2500	1 sample
2501-5000	1 additional sample

Initial control test sample should be taken at or near the start of production under each contract.

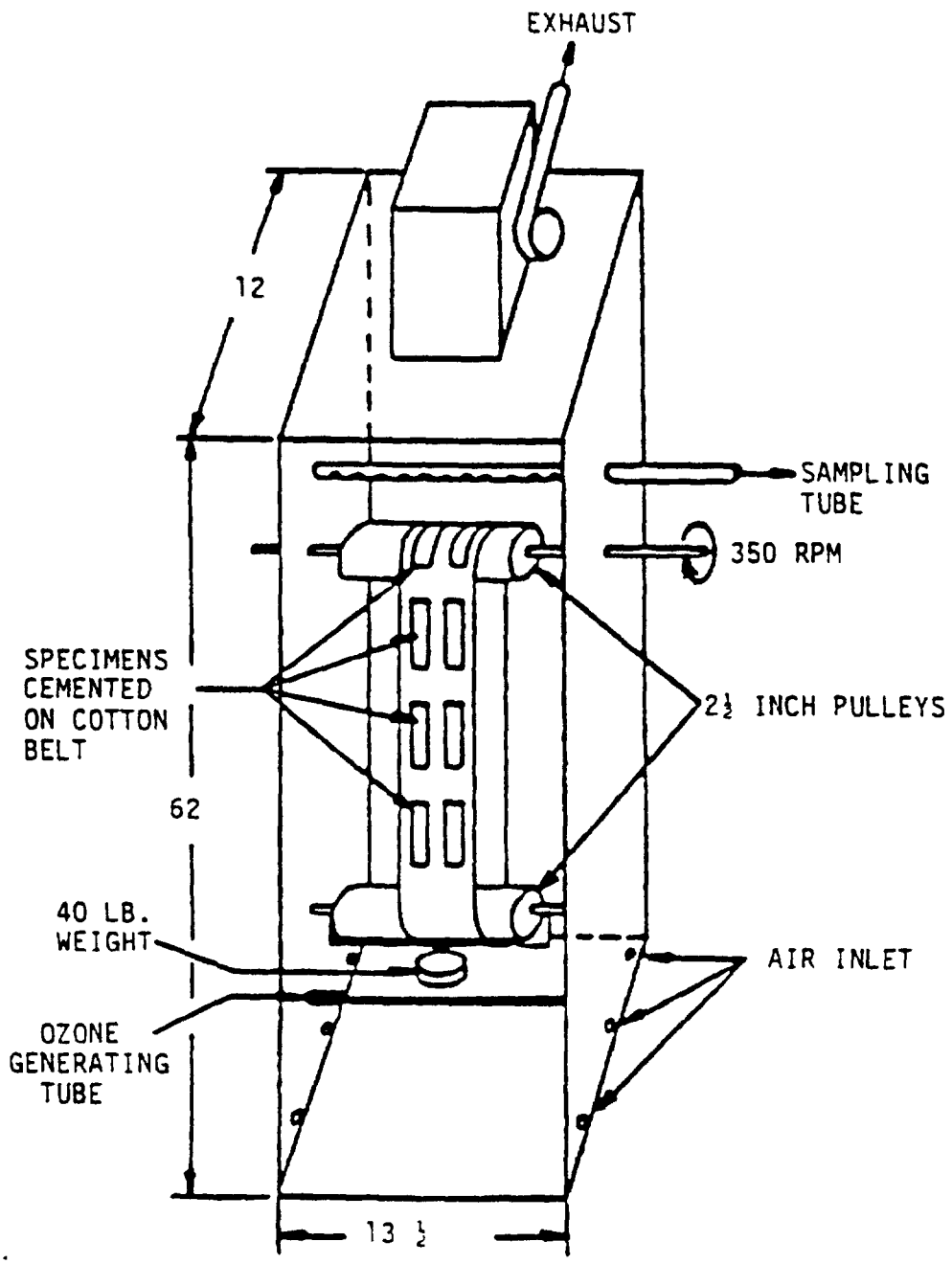
6.7.2.2 Applicable tests. The tires selected as specified in 6.7.2.1 should be subjected to the tests specified in table I, following examination for defects specified in table II.

6.8 Subject term (key word) listing.

Carcass, tread, sidewall, and bead.
Road Service/mission.
Temperature, ozone resistance

6.9 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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NOTE: Unless otherwise specified, dimensions are in inches.

FIGURE 1. Dynamic ozone testing apparatus.

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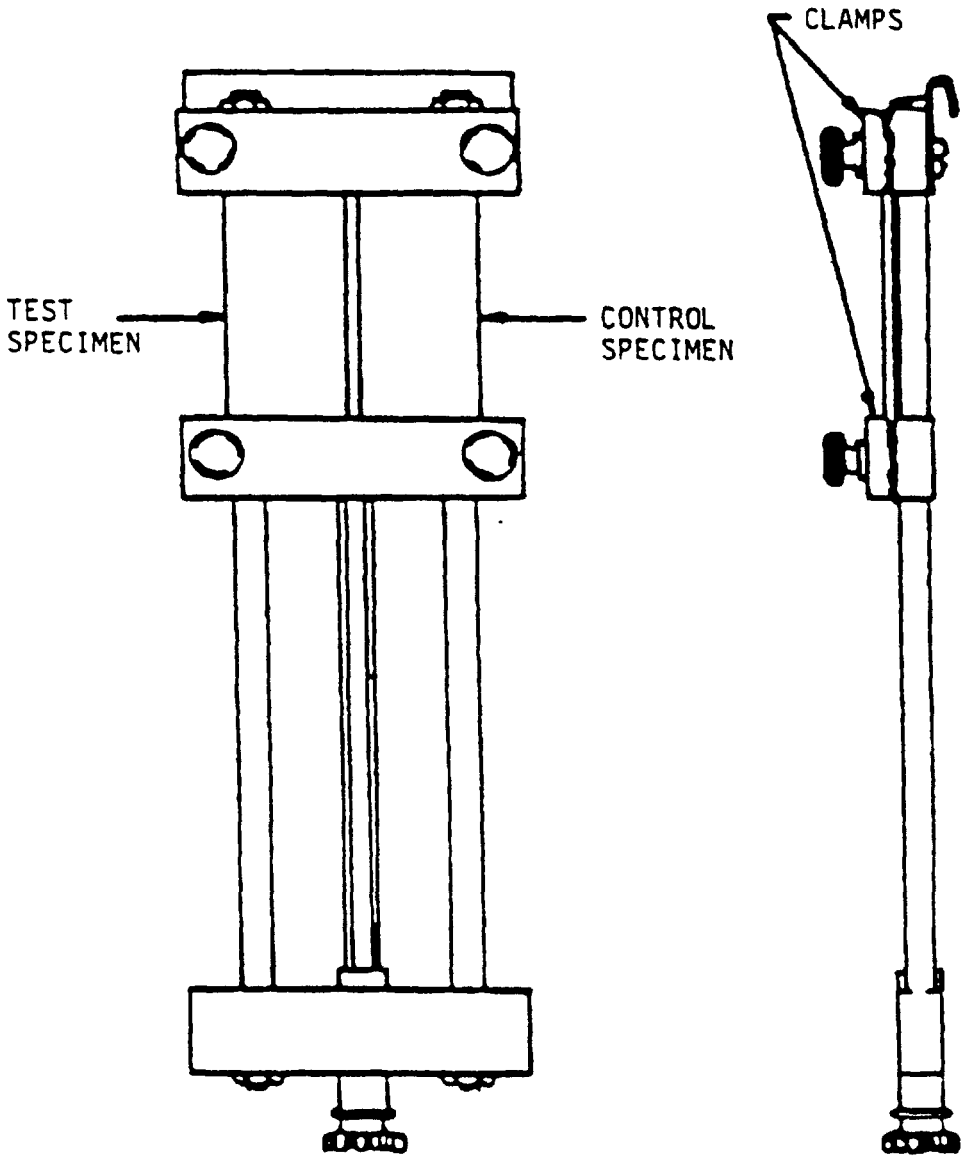


FIGURE 2. Specimen holder.

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